CORNELL HEDGE FUND

Mortgage-Backed Securities

Cornell Hedge Fund | NME

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Agenda

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1 Mortgage-Backed Securities Introduction

2 2008 Financial Crisis

3 Investment Thesis

4 Risks in 2008 in Comparison to 2022

5 Final Recommendation

Mortgage-Backed Security Introduction

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Mortgage-Backed Security (MBS) Overview

- An MBS is a type of bond that is backed by individual mortgages grouped together into one security
- Interest and principal payments on the MBS are secured by home and real estate loans
- The use of MBS were popularized by Lewis Ranieri, who is commonly known as the "father" of MBS

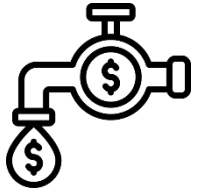
Unique Aspects of MBS

Diversification

Risk is decreased because the debt security is pooled together which **diversifies and lowers risk**

Increases Return

MBS offer higher yields than government bonds and higher coupon payments can increase return



Liquid Security

The **money received is used to offer other borrowers loans** which makes MBS a highly liquid security.

Potential Risks

1. **Increase in Interest Rate** → **Decrease in price of MBS**
2. **Default and Credit Risk** → **Investors Experience Loss**
3. **Negative Convexity** → **MBS price goes up less**

Advantages of investing in an MBS

- Cash flows are guaranteed by US government agencies
- Bond owners have a claim on the value of the asset in case of a default as opposed to solely losing their investment
- Investors are usually guaranteed a monthly payments of interest and principal

Sources: *WesternAsset.org*, *Investopedia*

Financial Crisis Primer: Profits Galore (1995-2006)

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Macro Review: Stock and Housing Markets

Stock Market

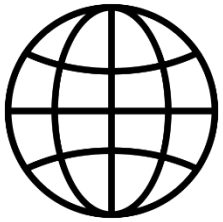
- Fueled by investor sentiment & expectation
- Inherent Risk-Return Reward

Housing/Real Estate Market

- Key indicator of overall economic health
- Increased wealth fuels economic investment, as financed via debt and equity markets

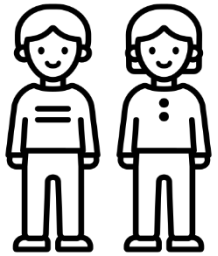
Market Conditioning

1995-2000:
Dot com Bubble



1999:
Gramm-Leach-Bliley Act

2004-2006:
Fed Policy & FNMA/FMCC



Sphere of Profitability: Mortgage Lenders, Institutional Investors, & Rating Agencies

Mortgage Lenders

-**Role:** Issue NINJA/Subprime ARM Loans
-**Assumed Hedge:** Increasing Home Prices
-**Benefit:** Risk Mitigation & Cash Flows

Institutional Investors

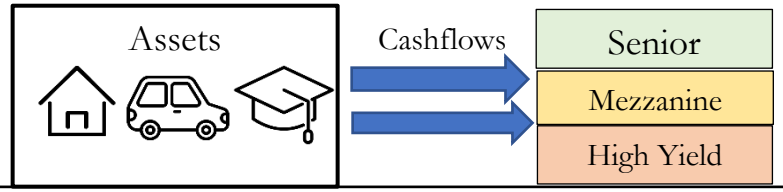
Role: Issue complex financial instruments to bolster cash flows and balance sheets.

Rating Agencies

Role: Give credence to risky securities issued by banks via AAA credit ratings.

Mortgage-Backed Securities

Collateralized Debt Obligations



Credit Default Swaps

Sources: *Principles of Macroeconomics, ColdFusion*

2008-2009 Financial Crisis: What Went Wrong

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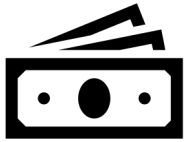
Domino Effect

2004 -2006	2007	March 2008	2008	September 2008
Federal Reserve raised interest rates from 1% to 5.25%	**2.3 Million Homes foreclosed by end of 2007**	**Bear Stearns collapsed**	**Goldman Sachs and Morgan Stanley bet against MBS**	**Lehman Brothers reached leverage ratio of 31:1**
ARM interest payments increased dramatically	Housing prices declined and mortgage-backed securities rapidly lost value globally	Bought by J.P. Morgan for $2/share, compared to value of $170/share earlier that year	Profited while investors and other banks experienced significant losses	The firm filed for the largest US bankruptcy of all time and were ultimately acquired by Barclays

Buyouts and Bailouts

$85 billion takeover of AIG



$700 billion Troubled Asset Relief Program

Public Protests

Lasting Effects

Fueled mass distrust in the government and financial institutions

Increased regulation of banking practices

"Too Big to Fail"

Sources: ColdFusion, Forbes

Investment Thesis

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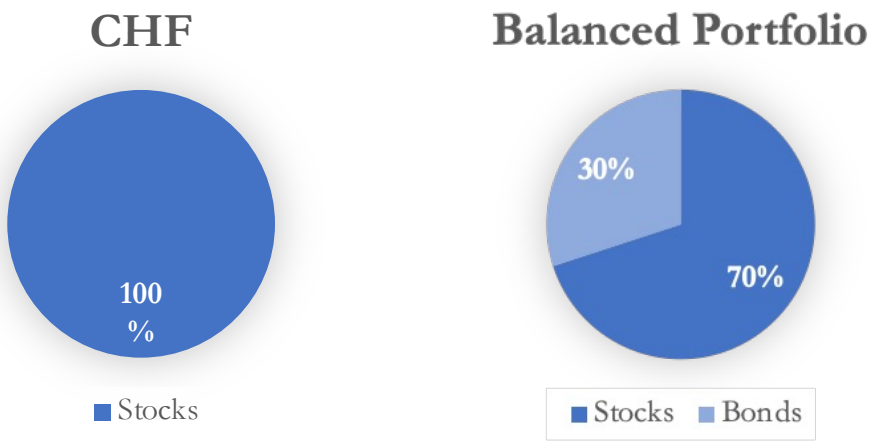
CHF should invest in long term mortgage-backed securities and reduce risk with a credit default swap.

Diversify Our Portfolio

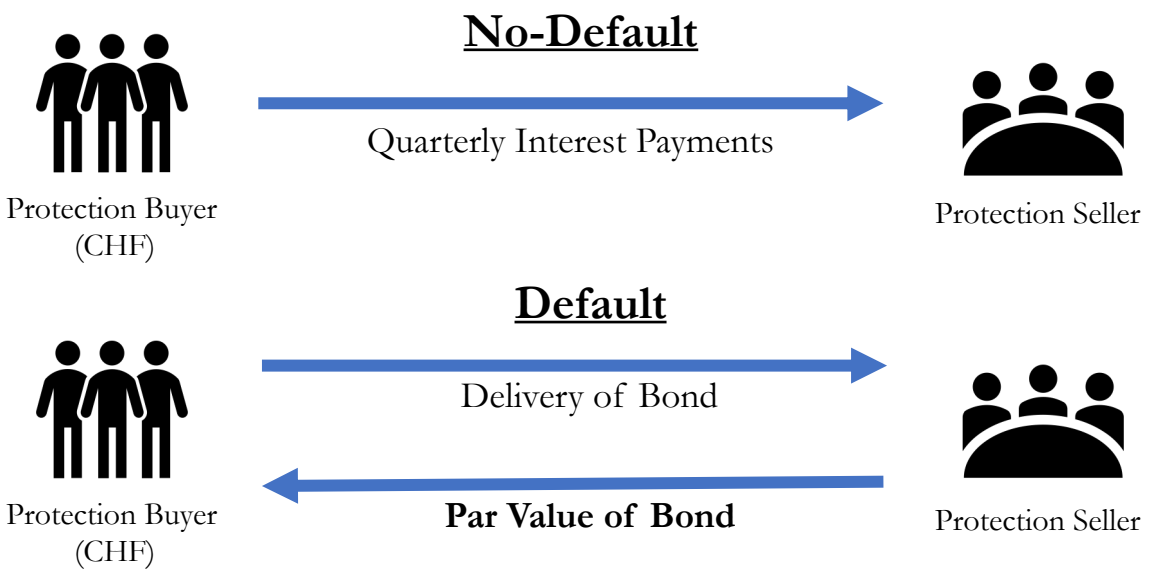


- Stocks are significantly more volatile than bonds; investing in both mitigates losses while still allowing for higher return
- MBS are also protected by assets and offer a diverse stream of cash flows

Credit Default Swap Mitigates Risk

Mortgage-Backed Securities always have a risk of defaulting, which can occur when the borrowers don't make their payments.

A Credit Default Swap functionally insures a party against payment defaults



Sources: Forbes, Insider, Seeking Alpha, ColdFusion YouTube

Investment Thesis: Increasing Demand for Housing

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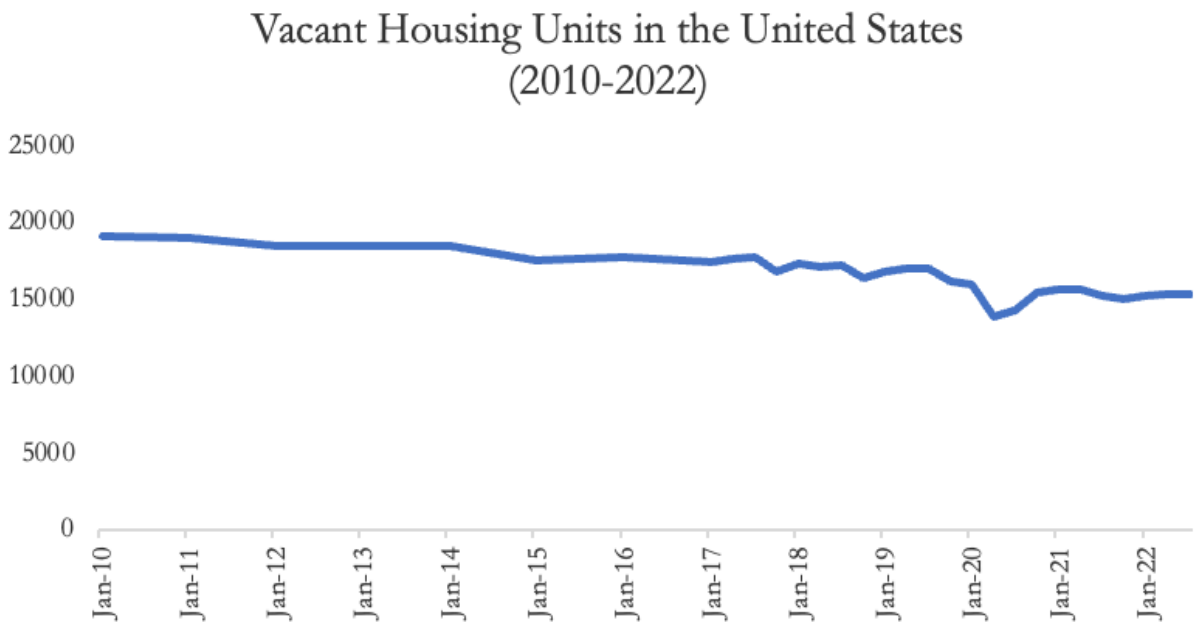
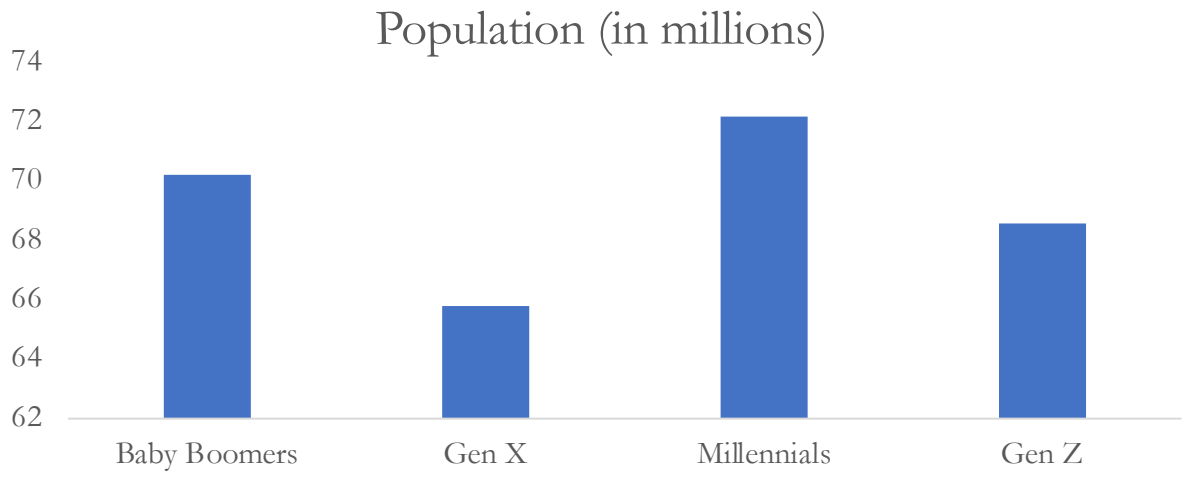
Decreasing Supply

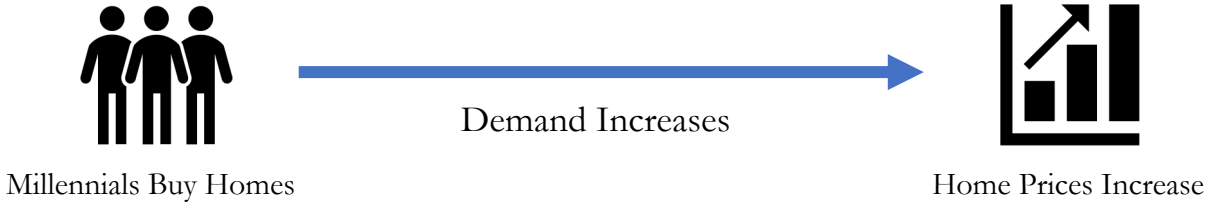


As available homes , the housing markets provides an increasingly reliable source for consistent cash flows.

Millennials Increase Demand



The average age of first-time home buyers is 33. Many millennials (ages 26-41) are approaching this step in their lives.



Sources: Morgan Stanley, NPR, Mortgage Solutions, Census.gov

Investment Thesis: Stricter Regulation of Bank Practices

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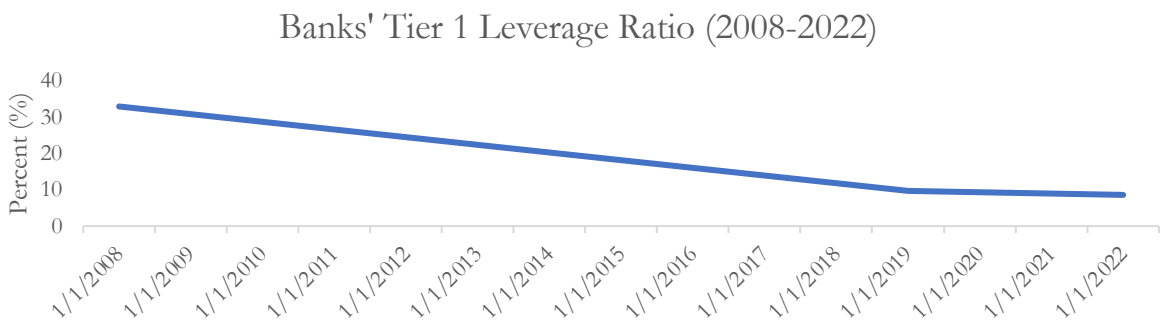
Newfound Regulation

Dodd-Frank Act:

Introduced in 2010 by Obama with the intention of putting more stringent checks and balances on the finance industry.

- Consumer Financial Protection Bureau (CFPB) was created, which ensures that banks and lenders do not predate on borrowers or engage in risky lending practices
- Office of Credit Ratings ensures credit agencies provide accurate credit ratings
- Whistleblower Program which can give anywhere from 10-30% of the proceeds from a litigation settlement



Minimum Credit Scores

Employment History

Debt-to-Income Ratio Limitations

Mandated Training of Mortgage Originators

Volcker Rule:

- Prohibits banks from using their own accounts for short-term proprietary trading and limits their dealings with covered funds
- It also prevents banks from using consumer funds to complete risky trades
- June 25, 2020, FDIC announced they will loosen restrictions on the Volcker rule, allowing banks slightly more freedom to invest

Sources: Bankregdata.org, Forbes

Investment Thesis: Macroeconomic Tailwinds

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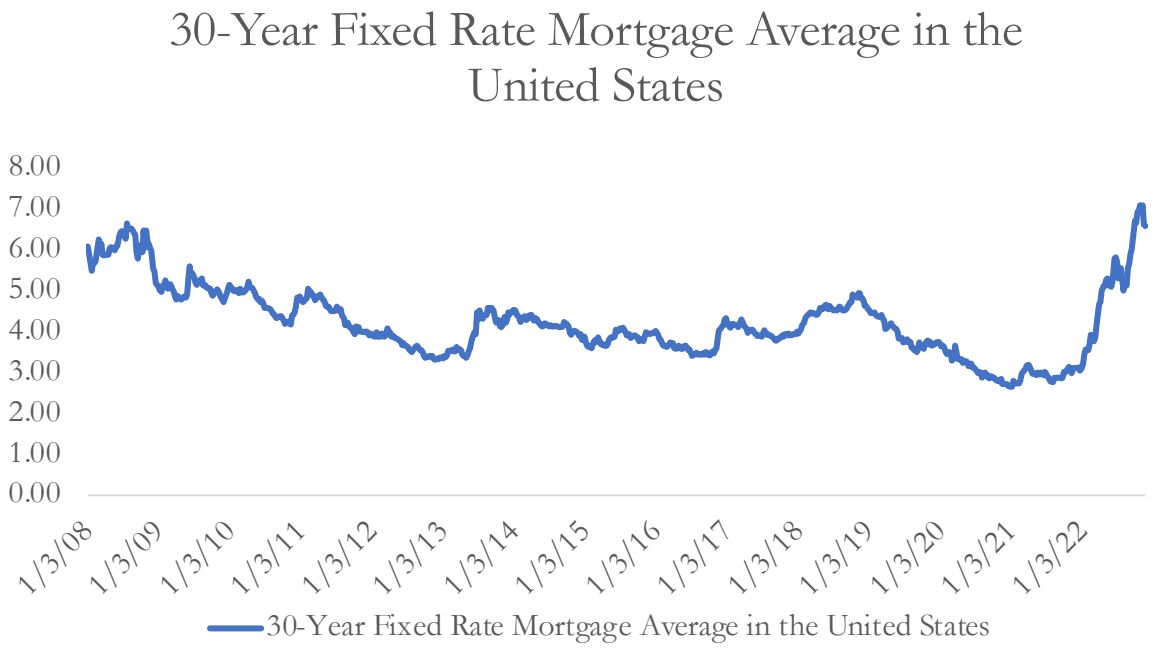
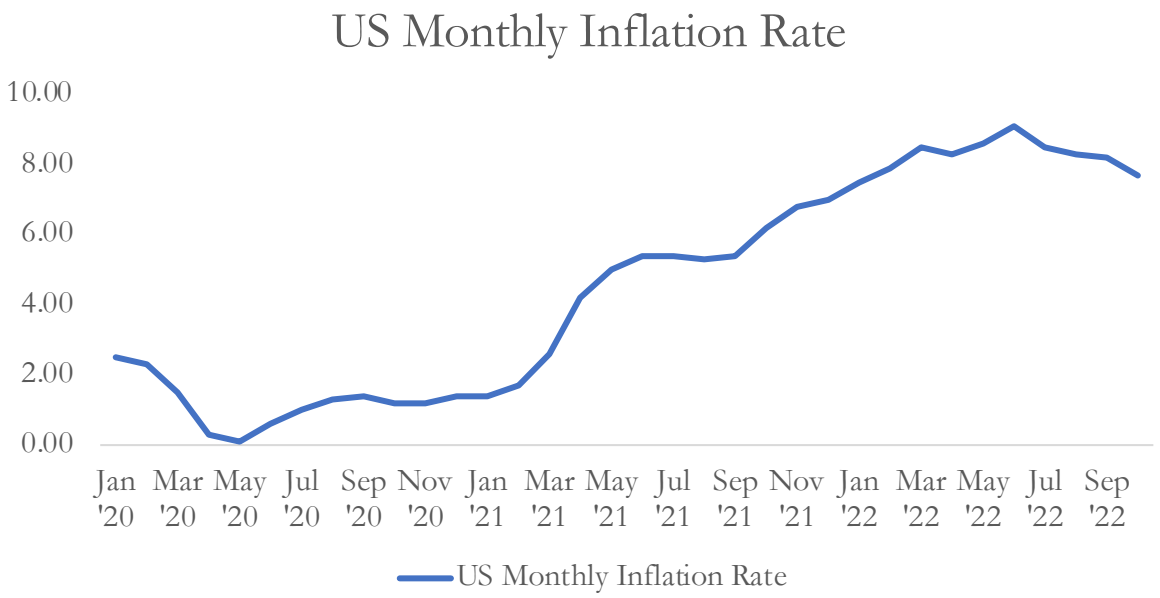
Increased Mortgage Rates Discourage Refinancing

Growing economic uncertainty and rising inflation caused the Federal Reserve to raise interest rates, which in turn, caused average mortgage rates to reach a 20-year high in early November.



Decreased Purchasing Power of Dollar Mitigates Risk of Prepayments

Inflation reached a four-decade high of 9.1% this year, decreasing consumers' purchasing power. In an uncertain economic environment, individuals will prioritize immediate needs and are less likely to prepay mortgages.



Sources: Forbes, Wall Street Journal, Federal Reserve Economic Data, Statista

Investment Thesis: Comparison of Risk

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2008

Sub-Prime Loans

The inclusion of sub-prime loans in MBSs was a catalyst in the 2008 financial crisis. When homeowners were unable to pay their mortgages, MBSs decreased in value.

Credit Agency Deceit

Credit rating agencies aimed to increase earnings and profits by giving inaccurately strong ratings to underperforming assets.

Falling U.S. House Prices

U.S. housing prices fell nearly 8.4% on average in early 2006, which decreased the value of mortgages and the resulting value of MBSs. They later fell 30% in 2009.

2022

Interest Rate Risk

When interest rates rise and are expected to continue rising, homeowners may refinance in expectation of increased rates. This leads MBS prices to decrease.

Prepayment Risk Pressures

Rising mortgage rates (currently 6.8%) can lead to prepayment risk – investors paying the principle ahead of schedule lowering value of MBS.

Recession

Recessions reduce economic activity and increase unemployment rates, which decreases housing prices and demand for mortgage financing, and increases default risk.

Sources: Rocket Mortgage, BankRate, The Balance

Final Recommendation

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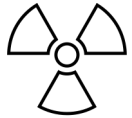
CHF should invest in Mortgage-Backed Securities in 6-9 months

Time Horizon

- According to Goldman Sachs, the U.S. economy will enter a recessionary period in 2Q23
- This recession is expected to last 11-15 months
- CHF should invest in MBS once macroeconomics effects that are currently acting as headwinds lessen in impact

Cornell Hedge Fund's Portfolio Risk

CHF can mitigate risk by investing in credit default swaps, which will act as insurance in the case of a volatile investment.



CHF is overexposed to risk related to equity investments. MBSs will lead to a more diversified portfolio.

Why are MBSs more attractive in 2022 vs 2008?

1 **Stricter Mortgage Standards**

Increased regulation regarding mortgages, including minimum credit scores, employment history, and debt-to-income ratio limits

2 **Credit Agency Reform**

The Credit Rating Agency Reform Act of 2006 regulates their internal processes, record-keeping, and business practices

3 **Wall Street Reform – Dodd Frank Act**

Introduced in 2010 by Obama with the intention of putting more stringent checks and balances on the finance industry.

Thank You